Real-Time Smart Zebra Crossing Safety Barrier System Using Computer Graphics

Submitted By

Thoushif Ahammed Farash Purno	Student- ID:0242310005101237

LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course

CSE412: Computer Graphics Lab in the Department of Computer Science and Engineering

DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision **Khandoker Nosiba Afrin,** Lecturer, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

Submitted To:

Khandoker Nosiba Afrin
Lecturer
Department of Computer Science and Engineering
Daffodil International University

Submitted by

Thoushif Ahammed Farash Purno
ID:0242310005101237
Section: 64-M(M2)
Dept. of CSE, DIU

COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO1	Design and implement fundamental computer graphics primitives and algorithms to construct 2D graphical scenes for real-life visualization problems having complex engineering attribute
CO2	Apply appropriate OpenGL programming techniques, geometric transformations, and graphics algorithms to solve complex real-world visualization problems using modern engineering tools.
CO3	Perform effectively as an individual and as a member of a diverse team to design, document, and implement computer graphics–based projects.
CO4	Create and present a complete computer graphics project by explaining complex engineering activities through effective reports, demonstrations, and presentations.

Mapping Course Outcome (COs) with the Teaching-Learning and Assessment Strategy

The mapping justification of this table is provided in section **4.2.1**, **4.2.2** and **4.2.3**.

CO	**POs**	**Teaching Learning Activities**	**Assessment Strategy**	**Learning Domains**	**Knowledge Profile**	**Complex Engineering Problem**	**Complex Engineering Activities**
CO1	PO1	TLA1 (Lab exercises)	Lab performance, Lab report	C2	K1, K3	EP1,EP4	
CO2	PO5	TLA2 (Algorithm implementation)	Lab performance, Lab report	C3, C4	K6	EP1, EP4, EP3	
CO3	PO9	TLA3 (Group project)	Lab project	P3, A2			
CO4	PO10	TLA4 (Presentation & Viva)	Project & Viva	C6, P3, A2			EA1, EA3

Table 2: Mapping of CO, PO, Blooms, KP and CEP

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Chapter 1

Introduction

1.1 Introduction

Computer Graphics is widely used to simulate real-life systems. In this project, we developed a Smart Zebra Crossing Safety Barrier System using OpenGL. The system visually represents a real-world traffic control mechanism where traffic lights control both vehicles and pedestrian safety barriers.
When the signal turns RED, vehicles stop and barriers appear (move upward) to protect pedestrians. When the signal turns GREEN, barriers move downward and vehicles start moving again.

1.2 Motivation.

Road accidents near zebra crossings are a serious issue, especially in school zones. Our motivation is to simulate a smart automated system that enhances pedestrian safety using computer graphics

1.3 Objectives

- Implement graphics algorithms (DDA, Bresenham, Midpoint Circle)
- Apply 2D transformations (translation, scaling)
- Design a real-time animated traffic system
- Synchronize **traffic signals, vehicle movement, and safety barriers**
- Demonstrate real-life safety solutions through graphics

1.4 Feasibility Study

Similar systems exist in modern smart cities where automated barriers and signals are used. However, most are hardware-based. Our project simulates this concept using software visualization (OpenGL).

1.5 Gap Analysis

Existing simulations:

- Show traffic lights only
- Do not include dynamic safety barriers + pedestrian logic

Our improvement:

1. Integrated barrier + vehicle + signal synchronization
2. Real-time animation

1.5 Project Outcome

The final output is an animated scene where:

- Traffic lights change automatically
- Barriers move up/down
- Vehicles stop/go accordingly
- Pedestrian safety is visually ensured

Chapter 2

System Architecture/ Scene Description & Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Scene Overview

The scene represents a **road with zebra crossing**, including:

- Traffic signal
- Moving car
- Pedestrian
- Safety barriers

2.1.2 Object Components

· Road (rectangles)
· Zebra crossing (parallel lines)
· Traffic light (circles)
· Car (rectangles + circles)
· Barriers (lines/rectangles)
· Pedestrian (simple shapes)

2.2 Use of Modern Tools

This section highlights the modern software tools and technologies employed in the Computer Graphics project to enhance visualization, animation control, and rendering performance. It explains how these tools assist in creating dynamic 2D scenes, debugging visual elements, and improving development efficiency.

✔ OpenGL / FreeGLUT – Open-source graphics libraries used for rendering 2D and 3D objects, applying transformations, and implementing real-time animations.

✔ Code::Blocks / Visual Studio – Integrated Development Environments (IDEs) used for writing, compiling, and debugging OpenGL programs

efficiently.
✔ GLUT / GLFW Libraries – Frameworks that manage window creation, input handling, and animation timing for smoother real-time rendering.
✔ C++ Programming Language – Core programming language used for implementing graphics algorithms (DDA, Bresenham, Midpoint Circle) and modular code structure.
✔ GIMP / Photoshop / Inkscape (Optional) – Tools used for creating or editing textures, backgrounds, and digital assets used within the scene.
✔ Git / GitHub – Version control systems for managing source code, tracking changes, and collaborating effectively among team members.

2.3 **Algorithm Used**

- DDA Algorithm → Drawing zebra crossing lines
- Bresenham Algorithm → Efficient line drawing
- Midpoint Circle Algorithm → Traffic light circles & wheels

These algorithms ensure accurate and efficient rendering.

2.4 **2D Transformations Applied**

· Translation → Car movement (x-axis)
· Translation → Barrier movement (y-axis)
· Scaling → Object size adjustments

2.5 **Animation Logic**

```
Car movement:carX += 0.01;
Barrier movement:if(signal == RED)
   barrierY += 0.01;
else
barrierY -= 0.01;
Signal control:
signalState = (signalState + 1) % 3;
```

Chapter 3

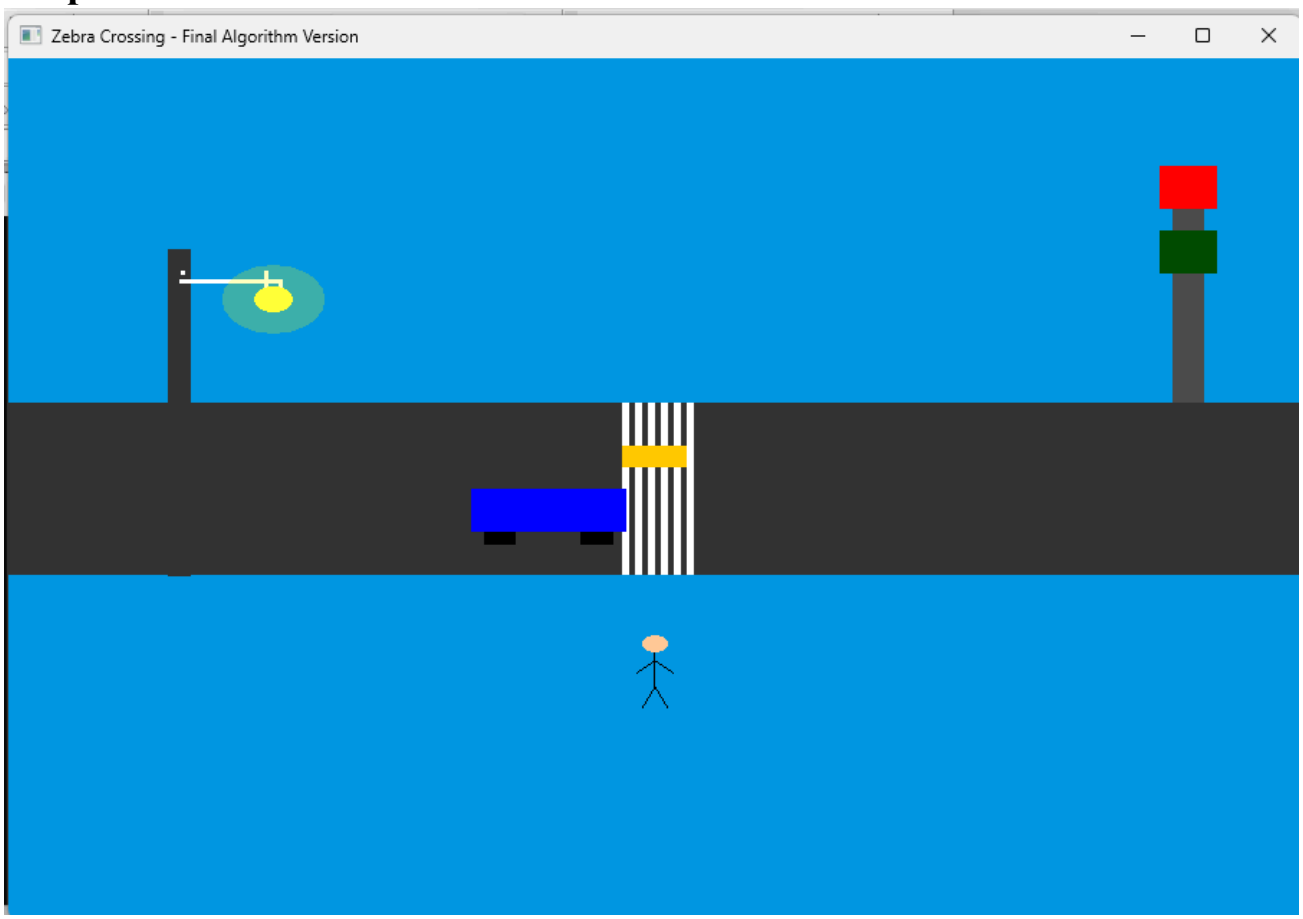
Implementation and Results

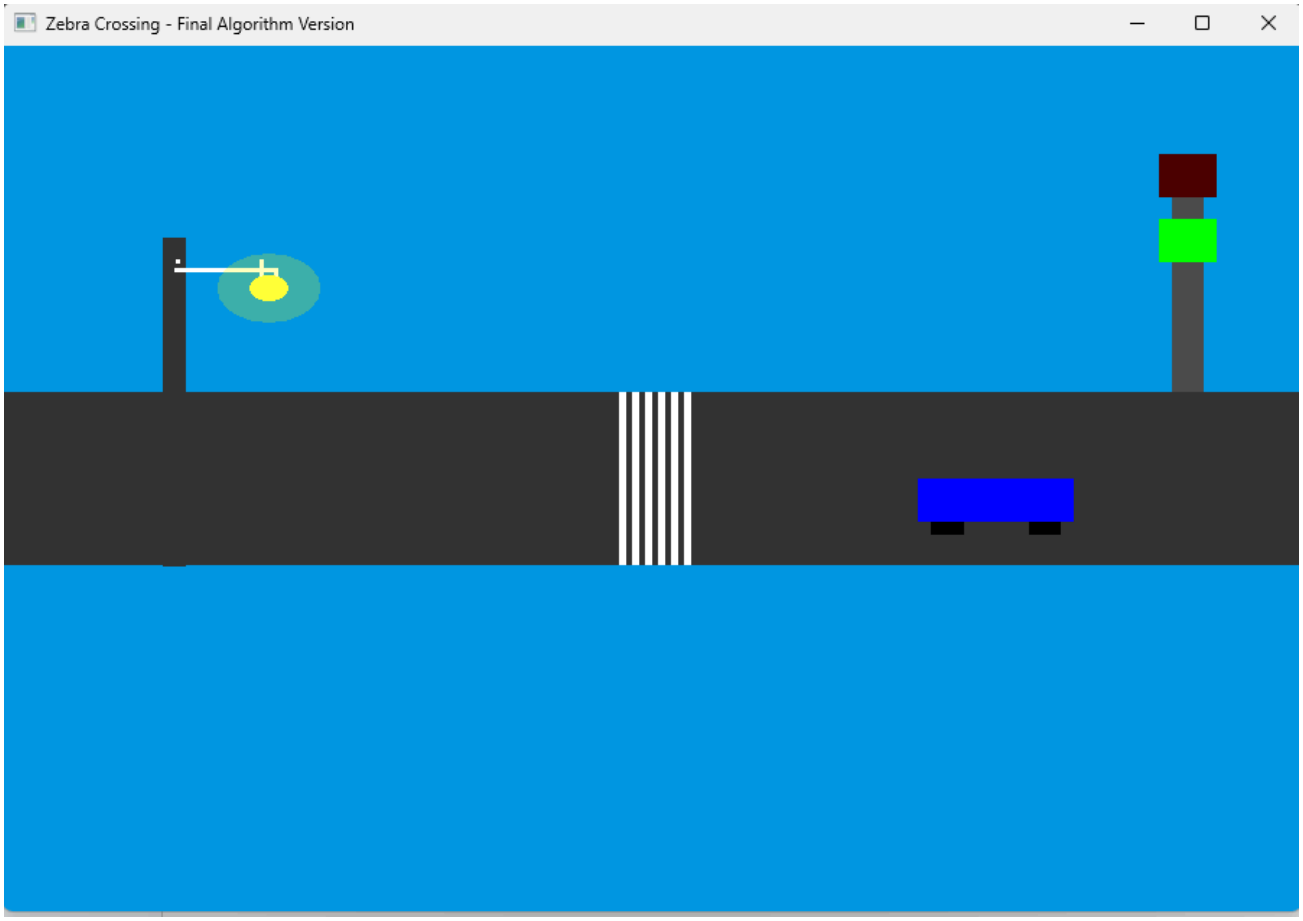
3.1 Implementation

The project is implemented using OpenGL where:

- Scene is drawn using primitive shapes
- Algorithms are used for accurate drawing
- Timer function controls animation

3.2 Output





3.3 Discussion

The system successfully demonstrates:
1. Real-time synchronization
2. Safe pedestrian crossing
3. Practical use of computer graphics

Chapter 4

Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

Improves pedestrian safety by ensuring that vehicles stop at the right time, allowing people to cross the road without risk. It reduces accidents by controlling traffic flow through signals and barriers, minimizing human error and careless driving. This system is especially helpful in busy areas like school zones, where children and pedestrians are more vulnerable. Overall, it creates a safer and more organized road environment for everyone.

4.1.2 Impact on Society & Environment

Encourages the development and adoption of smart traffic management systems that use automation and real-time control to improve road safety. It promotes urban safety solutions by integrating signals, sensors, and barriers to protect pedestrians, especially in high-risk areas like school zones and busy intersections.

From a societal perspective, it raises awareness about responsible driving and pedestrian rights, helping to build a more disciplined and safety-conscious community. It also supports city planning initiatives aimed at reducing congestion and improving overall transportation efficiency.

Environmentally, smoother traffic flow reduces unnecessary stopping and idling of vehicles, which helps lower fuel consumption and air pollution. This contributes to cleaner air and a healthier urban environment.

4.1.3 Ethical Aspects

- Original implementation
- No plagiarism
- Safe and meaningful design

4.1.4 Sustainability Plan

Can be extended into real-world IoT-based smart systems by integrating sensors, cameras, and connected traffic signals for real-time monitoring and control. These systems can automatically adjust traffic flow based on vehicle and pedestrian movement, improving efficiency. Using cloud data and analytics, authorities can predict traffic patterns and enhance long-term urban planning. Overall, it supports sustainable city development by reducing congestion, energy use, and environmental impact.

4.2 Complex Engineering Problem

4.2.1 Mapping of Program Outcome

Table 4.1: Justification of Program Outcomes

POs	Justification of Mapping (Project Perspective)
PO1	Mathematical & graphical logic
PO5	Use of OpenGL & tools
PO9	Solo work
PO10	Reporting & presentation

4.2.2 Complex Problem Solving

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Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	**EP2** Range of Conflicting Requirements	**EP3** Depth of Analysis	**EP4** Familiarity of Issues	**EP5** Extent of Applicable Codes	**EP6** Extent Of Stakeholder Involvement	**EP7** Inter-dependence
✓		✓	✓			

EP1: Synchronizing multiple moving objects
EP3: Managing animation timing
EP4: Handling real-time interaction

4.2.3 Complex Engineering Activities

Table 4.3 presents the mapping of the project with the Complex Engineering Activities (CEA) attributes. The attributes relevant to this project are marked (✓).

Table 4.3: Mapping with complex engineering activities.

EA1	EA2	EA3	EA4	EA5
✓		✓		

EA1: Scene design

EA3:

- Algorithm implementation
- Animation integration

Chapter 5

Conclusion

5.1 Summary

5.2

This project demonstrates a smart zebra crossing safety system using computer graphics. It integrates traffic lights, barriers, and vehicles to simulate real-world safety mechanisms.

5.3 Limitation

- No real sensor input
- Limited to 2D graphics
- No AI-based decision system

5.4 Future Work

- Add sensor-based pedestrian detection
- Upgrade to 3D simulation
- Integrate AI traffic control system
- Convert into real hardware system

References

Github: https://github.com/FarashBhai/Computer-Graphics-Project

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